

Safeguarding IoT networks against DDoS attacks using deep learning based zero trust network access

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Here, a deep learning-based zero trust network access (DL-ZTNA) system to enhance the security of the Message Queuing Telemetry Transport (MQTT) protocol within Internet of Things (IoT) applications was proposed. Combining multi-head convolutional neural networks and attention-based bi-directional long short-term memory networks with ZTNA provides real-time security analysis of device behaviour. This behaviour-based approach ensures that only authorized devices can access network resources and continuously monitors for potential threats like distributed denial of service (DDoS) attacks. The proposed DL-ZTNA system revokes device access when a threat is detected and prevents further malicious activities. Evaluation in a testbed environment showed improvements in CPU usage efficiency, throughput, and attack detection probability compared to traditional methods. This highlights the system's effectiveness in securing MQTT-based IoT networks against DDoS attacks while maintaining high performance, showcasing the potential of integrating deep learning techniques into ZTNA system for addressing security challenges in IoT environments.

Introduction: The increasing use of Internet of Things (IoT) devices in healthcare, smart homes, and industrial automation has raised concerns about their security. While IoT devices offer convenience, they are also prone to security vulnerabilities. Common IoT communication protocols like Message Queuing Telemetry Transport (MQTT) lack robust security features by default, making them susceptible to attacks. Additionally, the limited computational capabilities of IoT devices hinder the implementation of strong encryption and authentication methods, leaving them vulnerable to security threats such as distributed denial of service (DDoS) attacks and unauthorized access [1, 2]. In addition, recently, DDoS attacks have posed a significant threat to IoT networks by overwhelming devices and services with excessive traffic, disrupting normal operations. Due to the limited resources of IoT devices, mitigating DDoS attacks is particularly challenging, as traditional defence mechanisms often consume too much computational power, leaving devices vulnerable to attacks.

Zero trust network access (ZTNA) is a security framework that continuously verifies and authenticates devices accessing network resources to mitigate DDoS attacks. Unlike traditional security models, ZTNA does not automatically trust any device or user inside the network. Instead, it continuously verifies and authenticates all devices attempting to access network resources. Similarly, Software Defined Perimeter (SDP) method enhances security by creating an invisible barrier around network resources, allowing access only to authenticated devices and providing robust defence mechanisms against various network-based attacks. Other techniques, such as Identity and Access Management (IAM) and Micro-Segmentation (MSG), focus on managing and restricting access to network resources based on granular access controls [3].

The proposed deep learning-based zero trust network access (DL-ZTNA) system leverages deep learning to mitigate DDoS attacks on IoT networks [4]. In addition, the proposed DL-ZTNA system is specifically designed for IoT networks, where resource-constrained devices and high scalability pose unique security challenges. The system's real-time traffic analysis, combined with its lightweight deep learning models, ensures efficient detection of threats without overwhelming device resources, making it particularly suitable for securing IoT environments. Further, integrating ZTNA with deep learning enhances security and optimizes network performance [5]. Overall, the proposed system addresses key IoT security challenges by integrating deep learning with zero trust frameworks, enhancing DDoS protection, CPU efficiency, and detection accuracy, surpassing traditional methods like ZTNA and SDP.

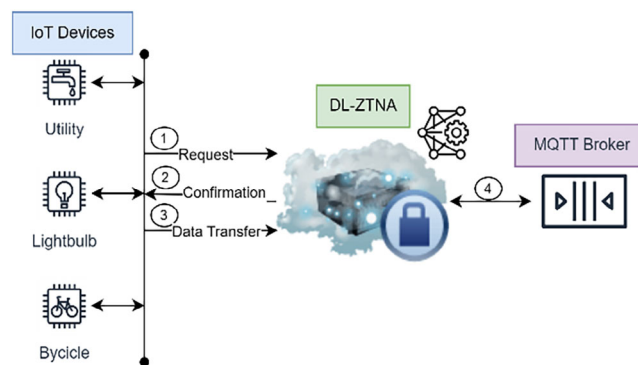


Fig. 1 The working overview of the deep learning-based zero trust network access testbed system

Zero trust network access: ZTNA is a security framework that operates on the principle of 'never trust, always verify'. It requires verifying every user, device, and application attempting to access resources, regardless of location. ZTNA enforces strict access controls based on the identity and context of the requesting entity to minimize the risk of unauthorized access and lateral movement within the network. When combined with deep learning (DL) techniques, ZTNA can analyse large amounts of data in real time, adapt to evolving threat landscapes, and provide a robust defence against sophisticated attacks. Implementing ZTNA involves several key phases to ensure a comprehensive security posture, especially in vulnerable IoT networks. These phases are discussed as follows.

- **User and device authentication:** Verifying the identity of users and devices before granting access.
- **Policy enforcement:** Applying security policies to determine what resources can be accessed.
- **Continuous monitoring:** The ZTNA monitors access and behaviour in real time, ensuring any anomalies are detected and addressed immediately. This continuous monitoring provides a sense of security, knowing that the system is always vigilant and ready to respond to potential threats.
- **Dynamic risk assessment:** Continuously assessing risk based on user behaviour and environmental factors.
- **Adaptive access control:** Adjusting access permissions in response to real-time risk assessments.
- **Integrating ZTNA with MQTT protocol:** We integrate ZTNA with the MQTT protocol by implementing a security framework that continuously authenticates and authorizes every device and user attempting to connect to the MQTT broker. The following section uses a step-by-step approach to achieve this integration.
- **Zero trust authentication before MQTT connection:** Before connecting to the MQTT broker, the end device must undergo a zero trust authentication process to verify its identity, security compliance, and context-based factors, ensuring that only authenticated and authorized devices can connect.
- **Enhanced security on TLS handshake:** The ensemble attention Long Short-Term Memory (LSTM)-convolutional neural networks (CNN) model helps prevent DDoS attacks during the TLS handshake by identifying and discarding malicious packets before processing. It only allows packets that have passed rigorous ZTNA authentication and contain unique, valid security tokens to proceed with the handshake.
- **Real-time threat detection:** The ensemble attention LSTM-CNN model monitors connected devices in real time. If it detects anomalies or potential threats, the ZTNA system revokes device access or isolates the threat to prevent DDoS attacks.

Finally, Figure 1 illustrates the high-level view of the proposed DL-ZTNA.

The working of the proposed system is depicted in Algorithm 1. Algorithm 1 begins with the initialization of the security status (S_d) and attempts counter (attempts(d)) for each device in the network. Devices undergo validation to check for a valid certificate and security compliance such as passing sets the security status to true; failing sets it to false. Subsequent actions increase the attempt counter. If the number of

Algorithm 1 Device authentication and distributed denial of service mitigation in deep learning-based zero trust network access-integrated Internet of Things networks

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1:  $S_d \leftarrow \text{false}$  for all  $d \in D$ 
2:  $\text{attempts}(d) \leftarrow 0$  for all  $d \in D$ 
3: for each device  $d \in D$ :
4:   if  $\text{valid\_cert}(d)$  and  $\text{security\_compliant}(d)$ :
5:      $S_d \leftarrow \text{true}$ 
6:   else:
7:      $S_d \leftarrow \text{false}$ 
8:   Increment  $\text{attempts}(d)$  for every publish/subscribe action
9:   if  $\text{attempts}(d) > A_{\max}$ :
10:     $S_d \leftarrow \text{false}$ 
11:    Log: "Access revoked due to excessive attempts"
12:    Continue to the next device
13:   else: if  $\text{anomaly\_score}(B_d) > \tau_d$ :
14:     $S_d \leftarrow \text{false}$ 
15:    Log: "Access revoked due to anomaly detection"
16:    Continue to the next device
17:   if  $M(B_d) > 0.5$ :
18:    Mitigate DDoS for  $d$ 
19:     $S_d \leftarrow \text{false}$ 
20:    Log: "DDoS attack detected and mitigated"
21:   end for
22: return  $S_d$  for all  $d \in D$ 

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attempts exceeds a predefined maximum threshold ($A_{\max}$), the device's access is revoked, and an entry is logged. Similarly, if the device's behaviour score ($\text{anomaly_score}(B_d)$) surpasses a certain threshold (τ_d), the device is flagged for suspicious activity, and its access is revoked. Suppose the DL model ($M(B_d)$) predicts a DDoS attack with a probability greater than a predefined threshold. In that case, the system mitigates the attack and revokes the device's access, logging the incident accordingly. Finally, the algorithm returns the security status for all devices in the network, ensuring that only compliant and non-malicious devices maintain access to network resources.

Simulation and discussion: We implement the testbed to integrate the MQTT protocol with the proposed DL-ZTNA approach. The experiment was conducted on a testbed with four IoT devices (Device_A, Device_B, Device_C, and Device_D), interacting with an MQTT broker via the DL-ZTNA system. Devices A and D had valid certificates and were security compliant, while B and C were compromised. These devices were connected to a PC available at the edge of the IoT network. The simulation ran for 60 s, during which traffic and behaviour were continuously monitored by the DL-ZTNA system. Further, the proposed system DL-ZTNA uses an ensemble model that integrates a multi-head CNN with an attention-based LSTM, which was trained using the CICDataset 2017 [6], consisting of 170,366 entries and 79 features, including Flow Duration, Total Fwd Packets, and Fwd Packet Length Max. One of the main reasons for selecting this dataset is that it is well suited for DDoS detection due to its rich feature set, which captures various aspects of network traffic.

The MQTT protocol is protected by the DL-ZTNA system, which monitors device behaviour in real time. The ensemble model assesses the probability of a DDoS attack based on features extracted from the network traffic. The dataset includes 170,366 entries with 79 features such as Flow Duration, Total Fwd Packets, and Fwd Packet Length Max. These features capture crucial network traffic aspects, enabling accurate device behaviour analysis under DDoS attacks. Similarly, if a device's

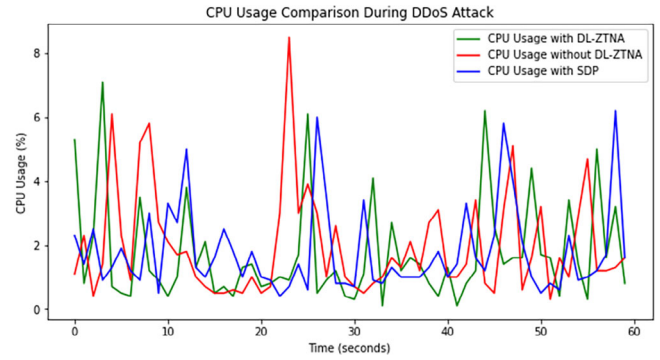


Fig. 2 Comparison of CPU usage during a distributed denial of service attack using deep learning-based zero trust network access (DL-ZTNA), non-DL-ZTNA, and Software Defined Perimeter methods

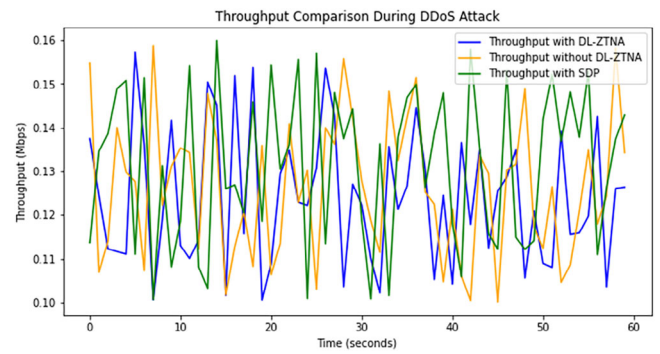


Fig. 3 Throughput comparison during distributed denial of service attack using deep learning-based zero trust network access, zero trust network access without deep learning, and Software Defined Perimeter methods

behaviour exceeds predefined thresholds, set to 100 in the simulation for suspicious activity, such as too many publish/subscribe attempts or a high anomaly score, the device's access is revoked. The simulation initializes the MQTT client, configures SSL settings, and registers devices. Further, the simulation runs for 60 s, during which the MQTT client processes traffic, and the DL-ZTNA system continually evaluates and mitigates potential DDoS attacks. Thresholds for suspicious behaviour were set at 100 publish/subscribe attempts, and any device exceeding this limit was flagged for anomaly detection. Similarly, devices exhibiting an anomaly score above the predefined threshold had their access revoked to mitigate potential threats. In the following section, we present several results obtained during the simulation. The results of the proposed DL-ZTNA are compared to simple ZTNA and SDP. As shown in Figure 2, the CPU usage during a DDoS attack is across three approaches: DL-ZTNA, non-DL-ZTNA, and SDP. The DL-ZTNA method shows moderate CPU usage, generally lower than the non-DL-ZTNA approach but slightly higher than the SDP method. This indicates that DL-ZTNA effectively balances security and performance, providing robust protection against DDoS attacks while maintaining reasonable resource usage, demonstrating its effectiveness in securing IoT networks.

As shown in Figure 3, the proposed DL-ZTNA method maintains more consistent throughput over time compared to the non-DL-ZTNA and SDP Methods, showcasing its effectiveness in handling DDoS attacks. The fluctuations in throughput are less severe in the DL-ZTNA method, indicating better network stability and resilience against attacks, making it a robust solution for securing IoT networks under DDoS conditions.

Figure 4 illustrates the attack detection probability over time for the proposed DL-ZTNA system compared to non-DL-ZTNA and SDP methods. It shows how the detection probability varies over 60 s during a simulated DDoS attack on an IoT network. The graph shows that the attack detection probability remains relatively high, fluctuating between approximately 0.5 and 1.0, with several peaks nearing 1.0. This indicates that the DL-ZTNA model accurately identifies potential threats during an attack, which is superior to the non-DL-ZTNA and SDP methods.

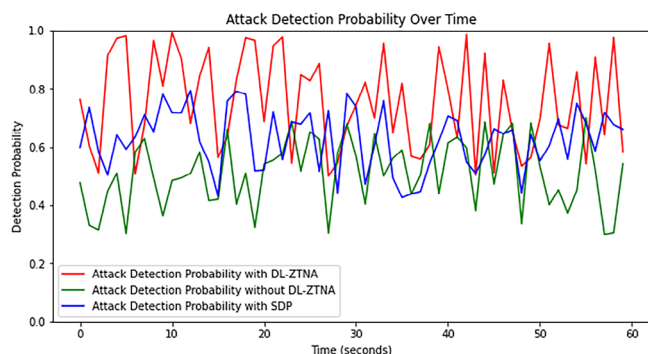


Fig. 4 Attack detection probability over time of the proposed deep learning-based zero trust network access (DL-ZTNA), non-DL-ZTNA, and Software Defined Perimeter methods

Conclusion: Traditional IoT security faces challenges due to inadequate authentication and MQTT vulnerabilities. The proposed DL-ZTNA system integrates deep learning models to analyse device behaviour in real time. It ensures that only authenticated devices access network resources, monitor activities, detect anomalies early, and revoke access promptly. Further, compared to traditional methods, DL-ZTNA reduces CPU usage, improves throughput, and enhances DDoS detection accuracy, making it a more efficient and robust solution for securing IoT networks while ensuring high performance. Future research will focus on enhancing DL-ZTNA's scalability, incorporating federated learning for reduced computational overhead, expanding support to other IoT protocols, and refining the model to handle emerging AI-driven cyberattacks.

Author contributions: **Murad Khan:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing—original draft; writing—review and editing.

Acknowledgements: This project was funded by the Kuwait Foundation for the Advancement of Sciences (KFAS) under project code: PN23-

15EM-1901. The author deeply acknowledges the Kuwait College of Science and Technology for supporting and providing a research environment to conduct this study.

Conflict of interest statement: The author declares no conflicts of interest.

Data availability statement: After the manuscript is formal accepted, the simulation code will be made available on the author's GitHub link. The link to the CICDataset 22017 used in the simulation is already provided in the manuscript's references list.

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Received: 28 August 2024 Accepted: 17 October 2024
doi: 10.1049/ell2.70075

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